ABDULLAH AHMED BHATTI

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PROFESSIONAL SUMMARY

Computer Engineering from Air University specializing in AI, computer vision, and hardware systems. Completed a Final Year Project building a deepfake detection system achieving over 90% accuracy using a multi-model ensemble (ViT-B/16, BiLSTM, Transformer Encoder). Hands-on background across VLSI design to deep learning pipelines. Seeking roles in applied AI, machine learning engineering, hardware/embedded systems, or software development where I can contribute technically from day one.

EDUCATION

B.Sc. Computer Engineering | Air University, Islamabad | CGPA: 3.04 *2022 – 2026*

- Graduated in June 2026
- Specialization in Machine Learning, Computer Vision, and Digital Hardware Design

Intermediate in Computer Science (ICS) | Punjab Group of Colleges, Islamabad *Completed 2022*

Matriculation | Silver Oaks School, Islamabad *Completed 2019*

EXPERIENCE

AI & Cybersecurity Intern | AgriStreams — Cybersecurity Zone, NSTP, NUST Islamabad *June – Aug 2025*

- Completed a structured internship at AgriStreams, in the Cybersecurity Department on network monitoring, security analysis, and vulnerability assessment tasks.
- Assisted in handling cybersecurity operations and gained practical exposure to enterprise security workflows and tools.
- Performed analysis related to Advanced Malware Detection (AMD) and security incident identification.
- Gained practical exposure to AI-based security solutions, cybersecurity frameworks, and applied research in an innovation-driven environment.
- Demonstrated strong teamwork, communication, and problem-solving skills during internship activities and technical assignments.

PROJECTS

1. Deepfake Detection in Multimedia Content Using Advanced Neural Models [Final Year Project - Completed]

- Built a multi-model ensemble deepfake detection system for video content, achieving over 90% accuracy across diverse real-world datasets.
- Temporal module: BiLSTM and Transformer Encoder for motion sequence and temporal inconsistency analysis.
- Spatial module: ViT-B/16 (Vision Transformer) for frame-level visual artifact detection.
- Leveraged GPU parallelism for accelerated inference while maintaining low computational overhead.
- Tech stack: PyTorch, OpenCV, Transformers, Python.

2. NLP Architecture Progression — Sentiment Classification

- Explored the evolution of NLP architectures from vanilla RNNs to LSTMs and Transformer-based models.
- Applied all three architectures to binary sentiment classification on the IMDb Movie Review dataset (positive/negative).
- Demonstrated performance improvements at each architectural stage, with Transformers achieving the highest accuracy.
- Tech stack: PyTorch, Python.

3. VLSI Design — Transmission Gate Full Adder
- Designed and implemented a full adder using Transmission Gate (TG) logic in a VLSI design project.
- TG-based design reduces transistor count and power consumption compared to standard CMOS implementations.
- Verified logic correctness through simulation and analysis of propagation delay and power metrics.
- Tools: Cadence design environment, Verilog.

4. 16-bit Processor Design — Verilog (Xilinx Vivado)
- Designed and implemented a fully parameterized 16-bit processor from scratch using Verilog HDL.
- Supported a custom instruction set architecture (ISA) with verified execution in Vivado simulation.

5. Image Processing Pipeline for Carotid Artery Ultrasound Images - Matlab
- Developed a MATLAB pipeline for Common Carotid Artery segmentation in ultrasound images using median filtering, contrast enhancement (CLAHE), and morphological operations, achieving ~0.84 Dice score.
- Implemented noise reduction, contrast enhancement, and automated segmentation on ultrasound images with quantitative evaluation using Dice Coefficient.

6. Bank Management System — OOP (C++)
- Designed and implemented a full bank management system in C++ using OOP principles.
- Incorporated ROT13 encryption and decryption for secure account data storage and retrieval.

8. MIPS Architecture Simulation — Assembly Language
- Simulated core MIPS processor architecture, demonstrating instruction-level operations and pipelining concepts.
- Implemented and tested programs in MIPS assembly, covering memory addressing, branching, and arithmetic operations.

TECHNICAL SKILLS

Machine Learning & AI: PyTorch, TensorFlow, OpenCV, Transformers, Pandas, NumPy, Scikit-learn, Deep Learning, Computer Vision, Image Processing, Whisper (Speech Recognition), Google Colab

Programming Languages: Python, C, C++, Assembly (MIPS), HTML, CSS

Hardware & VLSI: Verilog HDL (Xilinx Vivado), Cadence, Digital Logic Design, Transmission Gate Logic, VLSI Circuit Design, Microprocessors, Arduino, Proteus Simulation

Signal Processing & Robotics: Digital Signal Processing, Signals and Systems, Applied Robotics, MATLAB

Software Development: Object-Oriented Programming, Data Structures & Algorithms, Software Engineering Principles, Debugging, Problem Solving

Networking & Systems: CISCO Packet Tracer, Network Configuration and Troubleshooting, Ubuntu (Linux)

Databases: MS SQL Server

CERTIFICATIONS

- Microsoft Office Specialist (MOS) Certification — Microsoft

VOLUNTEER EXPERIENCE

- Volunteered at Pehli Kiran School providing academic support and mentorship to children from underprivileged backgrounds.
- Organised and participated in community ration drives and food distribution at Rehmani Free Dastarkhawan, Rawalpindi.
- Participated in a community cleanup initiative at F-9 Park, Islamabad.

LANGUAGES

English — Fluent **Urdu** — Native

REFERENCES

Available upon request.